

TrES-2b

R-0144

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_180614 · created 2026-07-16T08:08:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458283.6805 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.043
Transit depth (Rp/Rs)^2	2.19 +/- 1.28 [%]
Orbital Inclination (inc)	88.48 +/- 2.94
Airmass coefficient 1 (a1)	1.1 +/- 0.15
Airmass coefficient 2 (a2)	-0.07 +/- 0.12
Scatter in the residuals of the lightcurve fit is	12.97 %
Best Comparison Star	#2 - [237, 322]
Optimal Aperture	4.24
Optimal Annulus	7.33
Transit Duration (day)	0.061 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.043 vs 0.1254 (deviation 0.53σ)
Duration fit vs published	0.061 d vs 0.0746 d (deviation 1.23σ)
Mid-time O–C	1.4 min
Depth SNR	3.4
Residual scatter	12.97 %

Light curve

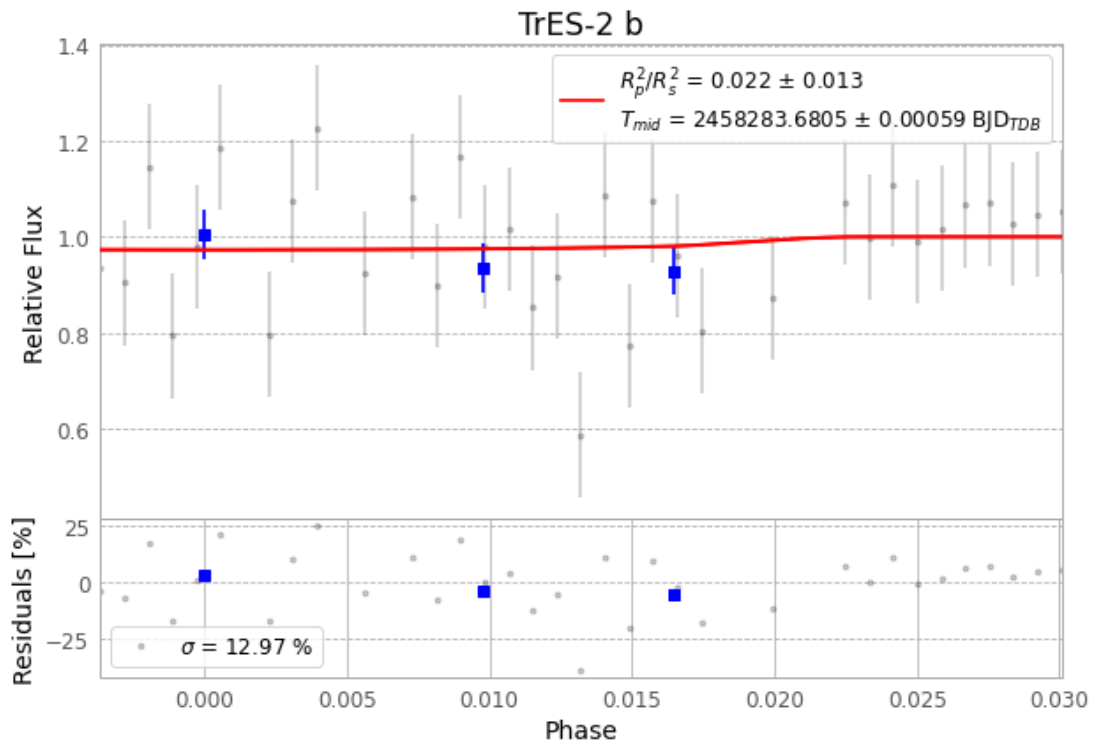


Figure 1 — FinalLightCurve_TrES-2 b_2018-06-13.png (SHA-256 7199bf20590caf73...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [376, 188], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08bbc65b43a1db02...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

41 FITS frames (27 MB), TRES-2180614040312.FITS → TRES-2180614060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180614.log (828839f21064...), FinalLightCurve_TrES-2 b_2018-06-13.png (7199bf20590c...), FinalLightCurve_TrES-2 b_2018-06-13.csv (e30c3823198b...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 08:08Z.